

Portfolio Flood Risk Assessment: Model Output

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1 Property-Level Outputs

1.1 Valuation Results

Field	Type	Description
property_id	string	Unique identifier
actual_value	float	Current value
predicted_value	float	Model prediction
value_difference	float	Prediction delta
value_difference_pct	float	Percentage delta

1.2 Risk Metrics

Field	Type	Description
flood_depth	float	Calculated depth
direct_impact	float	Value impact
cluster_id	integer	Risk cluster
local_density	float	Property density

2 Portfolio-Level Outputs

2.1 Aggregate Metrics

Metric	Type	Description
mean_impact	float	Average impact
var_95	float	95% VaR
es_95	float	95% ES
max_impact	float	Maximum impact

2.2 Concentration Analysis

Metric	Type	Description
geographic_concentration	float	HHI index
impact_concentration	float	Risk HHI
cluster_summary	dict	Cluster stats

3 Integration Output

These outputs serve as inputs for:

- Risk management systems
- Portfolio optimization models
- Regulatory reporting frameworks
- Insurance pricing models